

PARTS LIST

Semiconductors:

- IC1 - 7805 voltage regulator
- IC2 - LM35 temperature sensor
- SEN51 - SY-IHS-220 humidity sensor
- GS1 - MQ135 sensor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 4.7-kilo-ohm
- R2 - 100-ohm
- LDR1 - NORP 12 light-dependent resistor

Capacitors:

- C1 - 220uF, 16V electrolytic
- C2 - 4.7uF, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- CON2 - 4-pin connector
- TX1 - 1-pin connector
- JACK - DC jack
- LCD1 - 16x2 serial LCD
- Arduino Uno board

proximate values of the four sensors that are connected to the Arduino board. All the mentioned parameters are displayed on the LCD at an interval of 3 seconds.

Software

The software is written in Arduino programming language. The Arduino UNO is programmed using Arduino IDE software. ATmega328 on Arduino UNO comes with a boot loader that allows you to upload a new code to it without the use of external hardware programmer. It communicates using the STK500 protocol. You can also bypass the boot loader and program the microcontroller through ICSP (in-circuit serial programming) header, but using boot loader programming is quick and easy. Select the correct board from 'Tools->Board' menu in Arduino IDE and burn the program (sketch) through standard USB port in the computer.

Various operations are implemented as follows:

Temperature read. The ADC is read by using `analogRead()` instruction and corresponding ADC values are converted to temperature measurement as:

```
TempADCReading=analogRead
(TempSensorPin);
TemperatureCelsius=
(500.0*TempADCReading)/1024.0;
```

Humidity read. The response of humidity sensor is almost linear. Includ-

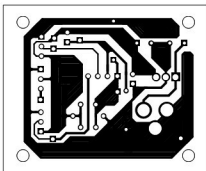


Fig. 6: An actual-size, single-side PCB for the environment monitoring system

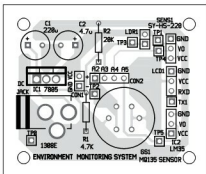


Fig. 7: Component layout for the PCB

ing the ADC resolution in the direct calculation of %RH we have used the below instructions to get the relative humidity:

```
HumidityADCReading=analogRead
(HumiditySensorPin);
RelativeHumidity=
(HumidityADCReading/6.7);
```

Light-intensity read. The output of the LDR is highly non-linear and exponential since the resistance decreases exponentially with the amount of light falling on the exposed surface. Hence to approximate it, you need to do piecewise linear or piecewise exponential approximation. The arrangement of the sensor is shown in the circuit diagram and the reading and calculating instructions are given below:

```
LightADCReading=analogRead
(LightSensorPin);
Vldr=(LightADCReading*4.88)/1000.0;
Rldr=((5.0*4.7)/Vldr)-4.7*1000.0;
b=log(1000000.0)-log(Rldr);
```

The 'Lux' value is calculated as below:

```
g=b/LDRconst;
LuxCalc=exp(g);
```

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

Air-quality read. This arrangement gives a rough approximation of the ppm levels of gas. These values are calculated as:

```
count=analogRead(GasSensorPin);
Vout=(count*4.88)/1000.0;
Rs=((5.0*R1)-(R1*Vout))/Vout;
ratio=Rs/approx;
GasConc=146.15*(2.868-ratio)+10;
ppm=GasConc;
```

Dew-point calculation using temperature sensor and humidity sensor. We have used the basic formula to calculate dew-point temperature using the data available from temperature sensor and humidity sensor. The instruction for calculating the dew point is:

```
z=(RelativeHumidity/100.0);
k=pow(z,0.125);
DewPoint=(k*(112.0+
(0.9*TemperatureCelsius))+
(0.1*TemperatureCelsius)-112.0);
```

Construction and testing

An actual-size, single-side PCB for the environment monitoring system is shown in Fig. 6 and its component layout in Fig. 7. Suitable connectors are provided on the PCB for input and output. Assemble the circuit on the PCB to minimise errors. Carefully assemble the components and double-check for any overlooked error. Connect the power supply through DC jack and provide power supply to the Arduino UNO board by connecting its Vcc and GND to CON1 on the PCB.

To test the circuit for proper functioning, check the correct power supply at TP1 with respect to TP0. The output of temperature, humidity, light and air-quality sensors can be checked at TP5, TP4, TP3 and TP2, respectively. In case the text is not visible on the LCD, adjust the potentiometer at the back of the serial LCD. ●

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